Financial Risk Modelling

Group project with StokTrak

Academic year 2025-2026

Introduction

The objective of this project is to implement modelling techniques to evaluate the risk of a portfolio of stocks. Thanks to the trading simulation platform StockTrak, you will have the opportunity to create your own portfolio and evaluate its risk. To do so, the project will be organized in 3 steps you can find in the table below.

N°	Step name	Description	Deadline
1	Group registration	Create your group and register to the platform	Before 24/10/2025
2	Portfolio construction	Select the stocks and create your portfolio	Before 06/11/2025
3	Risk modelling and evaluation	Derive forecasts for key risk measures of your portfolio	Before 19/12/2025

The output of the the project will be a report in which you justify and explain your modelling approach, derive and evaluate your forecasted risk measures. This report must be a pdf file you send to phubner@uliege.be for December 19$^{\text{th}}$, 2025, 11.59pm.

Procedure for group registration

Groups must be made of 4 members. To register, you send your group composition to phubner@uliege.be before October 24$^{\text{th}}$ with a designated team captain. This team captain will receive an email with the registration link to register to the StockTrak session. The access of the team captain to the StockTrak platform is the one of the team and should be shared accordingly. No other member of the team is allowed to register to the platform.

Instructions

Once you have your access to the platform, build your portfolio by placing trades. We require the portfolio to contain at least 6 different stocks. Beware that you have a limited cash amount available and that the amount not invested will be cash placed at 0%. The portfolio value and composition that you will have to analyse is the one asof November 6$^{\text{th}}$.

The objective of the analysis is to provide your best estimation of the Value-at-Risk (VaR) and Expected Shortfall (ES) of the portfolio at level 95%. You must use historical data prior to November 7$^{\text{th}}$ to predict the VaR and ES for the period going from November 7$^{\text{th}}$

to December 7th. After December 7th, you will receive the historical values of your portfolio on which you will perform a backtesting exercise.

The final report should include the following sections

1. Introduction
2. Data features: you analyse the key features of the risk factor, i.e. the log returns of the stocks in the portfolio. Refer to the different concepts seen in the course (heteroscedasticity, dependence aspect, extremes etc.).
3. Risk analysis: you describe the different approaches you use to derive VaR and ES values. Your analysis must meet the following criteria:
   - You must use and compare three different GARCH models.
   - You must use and compare three different copulas.
   - You must have three competing modelling approaches leading to three VaR and ES estimates.
   - Among the three modelling approaches, chose one to establish your forecast and provide quantitative and qualtitative justifications for your choice.
   - The forecast you provide is a daily forecast for each day between November 7th and December 7th.
4. Out-of-sample performance: you describe the results of your backtesting analysis on the realizations observed from November 7th to December 7th.

Data importation

StockTrak does not provide you with an easy access to historical data of the stocks you have in your portfolio. As such, you will have to download your own dataset. To do so, we provide you a R file called "MyGetSymbols.R", containing a function enabling you to import stock data directly from Yahoo Finance [1]. You are also provided with a file "myGetSymbols_example.R" with an example.

Evaluation

Each member of the teams that meet the instructions and hand a report that we consider of sufficient quality will be granted 1/20 bonus point at the exam. The team that we consider best in terms of report quality will be granted an additional 0.5/20 point at the final exam.

[1] Be aware that Yahoo Finance is not a database approved by the scientific community and thus cannot be used in the future for your master thesis.